

DETERMINISTIC PATH ENSEMBLES, DRIVEN COLLISION MAPS, AND ENDOGENOUS CONJUGATE PARAMETERS

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ABSTRACT. We construct a single finite-dimensional autonomous hybrid Hamiltonian realization of the driven baker benchmark and separate it from its symbolic natural extension. Parameter derivatives are defined on a fixed Sobolev-conormal Hilbert scale carrying bulk, face, corner, and higher flag currents. A closed all-order chain rule yields explicit response words with uniform remainder bounds. A Hilbert-valued Gordin decomposition then gives a Radon current-valued functional central limit theorem with trace-class covariance. Mechanical work and risk calibration remain distinct typed claims.

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1. ROUND-SEVENTEEN POSITIVE CLOSURE: A GLOBAL HYBRID HAMILTONIAN SYSTEM AND A CONORMAL RESPONSE CALCULUS

This section replaces the unmaterialized round-sixteen candidate. The mechanical realization is constructed globally before any response derivative is taken. Moving-seam derivatives are then defined in a fixed conormal Hilbert scale, not identified with ordinary Hölder cylinder functions. The

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current-valued invariance principle is proved by a Hilbert-valued martingale approximation.

1.1. A finite global return section. Let

$$0 = s_0(a) < s_1(a) < \cdots < s_4(a) = 1, \quad I_i(a) = (s_{i-1}(a), s_i(a)), \quad w_i = s_i - s_{i-1},$$

with a in a compact interval $K \Subset (-1, 1)$, so $w_i \geq w_* > 0$. On the branch $I_j(a) \times (0, 1)$, put

$$B_{a,j}(q, p) = \left(\frac{q - s_{j-1}}{w_j}, s_{j-1} + w_j p \right).$$

For $i, j, k \in \{1, 2, 3, 4\}$, define the open transition rectangle

$$S_{ijk}(a) = \{(q, p) : p \in I_i(a), q \in s_{j-1}(a) + w_j(a)I_k(a)\}.$$

The first index records the preceding strip, the second the active branch, and the third the next strip. We retain all one-sided faces as labelled faces. Thus the section

$$\widehat{S}_a = \bigsqcup_{i,j,k} \overline{S_{ijk}(a)}$$

is a finite compact manifold with corners; distinct germs are never identified on the section.

Lemma 1.1 (Correct exact primitive). *With $\lambda = p dq$,*

$$B_{a,j}^*(P dQ) - \lambda = dG_{a,j}, \quad G_{a,j}(q, p) = \frac{s_{j-1}(a)}{w_j(a)}(q - s_{j-1}(a)).$$

In particular there is no dp -term.

Proof. Since $dQ = dq/w_j$ for fixed a ,

$$B_{a,j}^*(P dQ) - p dq = \frac{s_{j-1} + w_j p}{w_j} dq - p dq = \frac{s_{j-1}}{w_j} dq = dG_{a,j}.$$

□

For every transition (i, j, k) , take a copy of $[0, 1]_t \times S_{ijk}(a) \times \mathbb{R}_E$ with symplectic form $dq \wedge dp + dt \wedge dE$. In three mutually disjoint clock windows use Hamiltonians generating translation by $-s_{j-1}$ in q , the linear flow of $-\log(w_j)qp$, and translation by s_{j-1} in p . At the outgoing face identify the point (Q, P) , with $Q \in I_k$ and $P \in I_j$, with the incoming sheet whose first two labels are (j, k) . The final label is determined after cutting Q into the four intervals $s_{k-1} + w_k I_\ell$. Gluings are therefore finite and are made only along closed exact Lagrangian graphs.

Theorem 1.2 (Global autonomous hybrid realization). *The preceding finite mapping-cylinder quotient is a Hausdorff, second-countable exact symplectic hybrid manifold $\mathcal{M}_a$ with corners. It carries one autonomous Hamiltonian flow on $\mathcal{M}_a \times T^*\mathbb{S}^1$ whose clock is strictly increasing, whose common section is $\widehat{S}_a$, and whose first-return relation is exactly the labelled baker return. On the regular full-measure set the return is single-valued, has return time*

one, has no parasitic returns, and projects to B_a on the ordinary square. Multiple-face encounters remain finite labelled hybrid states.

Proof. The local Hamiltonian flows are complete affine symplectic maps and their composition is $B_{a,j}$. Lemma 1.1 shows that each outgoing graph is exact. Source and target faces are compact and every gluing graph is closed; a finite quotient by such proper face identifications is Hausdorff and second countable. The clock Hamiltonian contains the term E , hence $\dot{t} = 1$. All channel perturbations are supported away from $t = 0, 1$, so an orbit meets the incoming section only at integer clock times. The finite source–target labels make the next sheet unique off the labelled seams. Exactness and the branch Jacobian one give preservation of the section area. At a corner the incident face label is part of the state, so no non-Hausdorff identification is introduced. $\square$

Proposition 1.3 (One global work pair). *After adjoining one canonical pair (z, π_z) , there is a smooth global Hamiltonian term for which*

$$z_{n+1} - z_n = c_j$$

on every channel with active branch j . It does not alter the return map on (q, p) .

Proof. Let $\rho \in C_c^\infty((0, 1))$ have integral one and add $\pi_z c_j \rho(t)$ on the disjoint channel labelled by j . These pieces are a single smooth function on the disjoint channel interiors and vanish in all gluing collars. Hamilton's equations give $\dot{z} = c_j \rho(t)$ and $\dot{\pi}_z = 0$; the other coordinates are unchanged by this term. $\square$

1.2. A fixed conormal Hilbert scale. Let $\mathcal{F}_n(a_0)$ be the finite stratification of the reference square by all forward and backward seam preimages of depth at most n . Every stratum is an affine rectangle, segment, or point. Choose smooth reference embeddings $\iota_{F,a}$ for $a \in K$, one-sided at a birth parameter. For codimension k , normal derivative order r , and depth n , put

$$\mathcal{H}_{n,k,r}^m = H^{-m-k-r}([0, 1]^2; N^{\otimes r}), \quad m > 4q + 8,$$

and define

$$\mathbb{J}_q^m = H^{-m}([0, 1]^2) \oplus \bigoplus_{n \geq 0} \bigoplus_{k+r \leq q} 2^{-n} (1+n)^{-2q-4} \mathcal{H}_{n,k,r}^m.$$

A family on a refinement is first summed as an ambient distribution and only then measured. Hence subdividing a face does not change its norm. The bulk component occurs once and the constant bulk density has its ordinary H^{-m} -norm.

For a smooth current T_F on a reference stratum define independently

$$J_{F,r}(a) = \partial_a^r(\iota_{F,a})_\# T_F$$

by differentiating its pairing with a smooth test form. Clean intersections of the affine vertical and horizontal strata define corner traces by the standard

iterated Sobolev trace; the choice $m > 4q + 8$ leaves more than one trace order at every step.

Theorem 1.4 (Closed conormal jet calculus). *For every fixed q :*

- (i) *all face restriction, boundary, and clean-intersection maps extend continuously between the components of $\mathbb{J}_q^m$;*
- (ii) *the material derivative $\mathfrak{D}_a$, first defined by the ambient distributional formula above, is closable from $\mathbb{J}_{q+1}^{m+2}$ to $\mathbb{J}_q^m$;*
- (iii) *the labelled return push-forward $\mathcal{T}_a$ and its parameter derivatives satisfy, uniformly on K ,*

$$\|\partial_a^r \mathcal{T}_a J\|_{\mathbb{J}_q^m} \leq C_q^{r+1} r! \|J\|_{\mathbb{J}_{q+r}^{m+2r}}, \quad 0 \leq r \leq q;$$

- (iv) *every repeated moving-boundary derivative is the finite clean conormal chain-rule sum over flags, is independent of local defining functions, and has a Taylor remainder bounded by*

$$\|\mathcal{T}_{a+h} J - \sum_{r=0}^q \frac{h^r}{r!} \partial_a^r \mathcal{T}_a J\|_{\mathbb{J}_0^m} \leq C_q |h|^{q+1} \|J\|_{\mathbb{J}_{q+1}^{m+2q+2}}.$$

Proof. The trace theorem gives a bounded map $H^s \rightarrow H^{s-1/2}$ on every affine face; duality gives the asserted maps on negative spaces. Vertical and horizontal faces meet cleanly, so iterated traces commute up to the orientation sign and no product of two unrelated distributions is formed. If $J_n \rightarrow 0$ in the stronger space and $\mathfrak{D}_a J_n \rightarrow K$ in the weaker space, pairing with a smooth form and integrating the scalar derivative over a parameter interval gives $\langle K, \phi \rangle = 0$; hence the derivative is closable.

Every branch map and every reference embedding is affine-rational in the parameters with denominator bounded below by w_* . The distributional Faà di Bruno formula therefore gives at most $C_q^r r!$ terms, each a bounded affine push-forward followed by at most r trace operations. A preimage can increase the depth by at most one; the factor 2^{-n} absorbs that shift, while summing ambient currents makes the estimate independent of how a face was subdivided. The same induction proves the flag chain rule. Taylor's integral remainder and the estimate at order $q+1$ give (iv). $\square$

1.3. Response words and the current-valued invariance principle.

Let L_a^+ and L_a^- be the one-sided Bernoulli transfer operators on future and past Hölder spaces and let $R_a^\pm = (I - L_a^\pm)^{-1} Q_a^\pm$. Let $S_a^{(r)}$ be the r -th score multiplier, $C_a^{(r)}$ the present-symbol coupling derivative, and $D_a^{(r)}$ the closed conormal jet operator. A response word of total order q is an ordered product in

$$D_a^{(r)}, \quad S_a^{(r)} R_a^+, \quad S_a^{(r)} R_a^-, \quad C_a^{(r)} R_a^+ R_a^-,$$

whose derivative orders sum to q . Its coefficient is the corresponding multivariate Bell coefficient.

Theorem 1.5 (Explicit all-order response). *For a C^{q+1} transported physical current $F_a \in \mathbb{J}_{q+1}^{m+2q+2}$,*

$$\frac{d^q}{da^q} \Lambda_a(F_a) = \sum_{W \in \mathfrak{W}_q} c(W) \Lambda_a(WF_a),$$

where every source and target space is determined by Theorem 1.4. Uniformly for $a \in K$,

$$\sum_{W \in \mathfrak{W}_q} |c(W)| \|\Lambda_a W\| \leq C_q (1 - \rho_K)^{-2q-2}.$$

The formula is invariant under dynamic refinement. If F is a fixed physical observable rather than a transported observable, the score, transport, and flag terms cancel and $d^q \int F dq dp / da^q = 0$.

Proof. Differentiate a finite past–future cylinder pairing. The ordinary Leibniz rule produces precisely a score, present coupling, or conormal material insertion. Centered terms are solved by the two one-sided Poisson equations, which insert $R_a^\pm$. Induction gives the ordered word set and the Bell coefficient. The one-sided gaps bound the resolvents; the factorial conormal bounds in Theorem 1.4 bound every remaining insertion. Absolute summability permits the cylinder depth to tend to infinity. Refinement invariance is ambient-current additivity. The last claim follows by differentiating the exact change-of-variables identity for Lebesgue area. $\square$

Fix q , and let $Y_j^{(q)}$ be the centered q -jet increment at the j -th return, transported to the reference space $\mathbb{J}_q^m$. The geometric depth weights and the Bernoulli cylinder weights imply

$$\sum_{n \geq 0} \|\mathbb{E}(Y_0^{(q)} \mid \mathcal{F}_{-n}^0) - \mathbb{E}(Y_0^{(q)} \mid \mathcal{F}_{-n+1}^0)\|_{L^2(\mathbb{J}_q^m)} < \infty.$$

Theorem 1.6 (Radon current-valued functional CLT). *The covariance series*

$$\Sigma_q = \sum_{k \in \mathbb{Z}} \mathbb{E}[Y_0^{(q)} \otimes Y_k^{(q)}]$$

converges in trace norm on the separable Hilbert space $\mathbb{J}_q^m$. Moreover

$$n^{-1/2} \sum_{j < nt} Y_j^{(q)} \implies W_q(t)$$

in $D([0, 1], \mathbb{J}_q^m)$, where W_q is a Radon Brownian motion with covariance Σ_q . The statement includes all bulk, face, corner, and higher-flag components of order at most q .

Proof. The displayed projective summability gives the Hilbert-valued Gordin decomposition $Y_j = M_j + H \circ \sigma^{j+1} - H \circ \sigma^j$, with (M_j) a stationary square-integrable martingale difference sequence. The conormal estimate and $w_i \geq w_*$ give $\mathbb{E}\|Y_0\|^2 < \infty$. Hence the covariance operator is positive and trace class, with trace bounded by the absolutely summable correlation series. The Hilbert-space martingale invariance principle gives convergence of the

martingale partial sums. The coboundary is negligible uniformly in time after division by $\sqrt{n}$. Burkholder’s inequality gives tightness; separability makes the Gaussian law Radon. $\square$

Theorem 1.7 (Separated calibration and A1 closure). *The benchmark admits one global autonomous hybrid Hamiltonian realization, one global canonical work coordinate, a closed all-order conormal response calculus, explicit summable response words, and a Radon current-valued FCLT. The mechanical conjugate field equals an entropic valuation coefficient only under the additional equality of their likelihood cocycles.*

Proof. The mechanical assertions are Theorem 1.2 and Proposition 1.3. The response and fluctuation assertions are Theorems 1.4, 1.5, and 1.6. Equality of the two scalar coefficients follows only if their Radon–Nikodym exponents agree on the nonconstant branch work; then the coefficients are equal. No such identification is made without that compatibility axiom. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Seventeen dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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